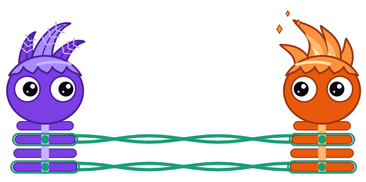
The final twins trained on 43 tasks in 42,832 episodes

Quill
Qwen3-4B



Ember
AFM-4.5B

		Percent of each stage's episodes			
		Warm-up 900 steps	**Full mix** 1,320 steps	**Chat training** 157 steps	**More chat training** 300 steps	**Episodes in all**
Value games	Write the partner's secret word	14.3%	4.7%	4.1%	3.1%	**3,306**
	Write the partner's secret number	14.2%	4.5%	4.5%	2.6%	**3,242**
	Write the partner's two-word phrase	14.7%	4.4%	3.0%	2.7%	**3,242**
	Write the partner's string of digits	14.3%	4.8%	4.3%	2.9%	**3,326**
	Add own number to the partner's	14.2%	4.6%	4.4%	3.1%	**3,280**
	Join a name and a city, one each	14.4%	4.7%	4.1%	3.0%	**3,305**
	Look up a code in the partner's table	14.0%	4.7%	3.7%	2.7%	**3,227**
Steering, identity, own answers	Make the partner mention a concept	4.7%	3.1%	2.6%	**1,186**
	Say who you are, or who the partner is	9.9%	8.7%	7.6%	**2,666**
	Answer as the original model did	9.6%	7.8%	7.3%	**2,563**
Dialogue games	Ask and answer questions in turn	1.1%	0.8%	0.6%	**277**
	Chat, then say who you are	1.0%	0.8%	0.6%	**268**
	Recall a word sent early in the chat	1.1%	0.8%	0.7%	**296**
	Learn a made-up fact, then take a quiz	1.1%	0.9%	0.7%	**288**
	Guess the partner's unspoken concept	1.0%	0.8%	0.6%	**268**
	Agree on one of two private concepts	1.1%	0.8%	0.6%	**289**
	Steer the partner while only listening	1.2%	0.7%	0.6%	**295**
	Write a word sent during an answer	0.9%	0.5%	0.6%	**228**
Shared pen	Write a poem together, a word each	2.2%	1.9%	1.2%	**564**
	End the poem on the partner's word	2.4%	2.0%	1.6%	**640**
Shared reading	Write out the text the partner reads	2.7%	1.8%	1.4%	**673**
	Each writes out the partner's text	2.7%	2.3%	1.8%	**708**
	Join two halves of a text, one each	1.7%	1.2%	0.9%	**420**
	Continue the text the partner reads	1.7%	1.4%	0.8%	**429**
	Reply as if reading the partner's text	5.4%	3.7%	3.0%	**1,382**
	Answer about the partner's text, then talk	4.4%	3.8%	2.5%	**1,148**
Tasks on the partner's text	Answer a question from the partner's text	1.2%	1.1%	0.5%	**312**
	Answer from two paragraphs, one each	0.7%	0.3%	0.3%	**170**
	Translate the partner's sentence	1.6%	1.2%	0.8%	**403**
	Paraphrase the partner's sentence	0.5%	0.4%	0.2%	**119**
	Say if one sentence implies the other	0.4%	0.4%	0.3%	**111**
	Solve the partner's word problem	0.4%	0.2%	0.3%	**98**
	Say what the partner's code returns	0.2%	0.2%	0.2%	**57**
	Write code for the partner's problem	0.8%	0.8%	0.3%	**206**
	Follow the partner's instruction	0.8%	0.7%	0.8%	**230**
	Join a question and options, one each	0.8%	0.5%	0.5%	**198**
Play	Guess the partner's secret word	0.7%	0.6%	0.4%	**185**
	Give the gift the partner wishes for	1.5%	1.3%	0.6%	**371**
	Say which city the partner is in	1.1%	1.4%	0.7%	**301**
	Put the partner's character in a story	0.3%	0.3%	0.2%	**87**
	End a rhyme on the partner's word	0.7%	0.6%	0.4%	**190**
Free chat	Chat with no game or task		18.2%	18.5%	**1,345**
	Chat when a twin has a private pet, job, ...			19.4%	**933**
Silent steps	Twins write 4 spaces before replies	47.9%	46.8%	47.7%	**13,584**